

Git Beginner Notes

1. Repository

- A repository is like a project folder that stores your code and history.
- Two types:
 - Local Repo – On your computer.
 - Remote Repo – On GitHub/GitLab/Azure DevOps.

Example:

```
git init
git clone <url>
```

2. Git Source Control / Version Control

- Git keeps track of changes in your code over time.
- You can go back to old versions, work in parallel with others, and merge later.

3. New Branch Creation from Existing Branch

- A branch is like a copy of your project where you can work without affecting others.

Example:

```
git checkout main
git pull origin main
git checkout -b feature1
```

4. Local Branch vs Remote Branch

- Local Branch → Exists only on your computer.
- Remote Branch → Exists on GitHub (or other server).

5. Pull Request (PR)

- A Pull Request is asking teammates to review your code before adding it to main.

Steps:

1. Push your branch to remote.
2. Open PR on GitHub/Azure DevOps.
3. Reviewer checks, comments, approves.
4. Merge into main.

6. Compare Code in Editor

- Compare your code with old code to see what changed.

Example:

```
git diff
```

7. How Many Files are Changed in Git

- Check changed files:

```
git status
```

- See how many changes in each file:

```
git diff --stat
```

8. Approval Process

- Before merging into main, someone must approve your PR.

Steps:

1. Developer creates PR.
2. Reviewer reads and approves.
3. If changes needed → send back.
4. After approval → merge branch.

9. Merge Conflicts (and How to Fix)

- A merge conflict happens when two people change the same line in a file.

- Git doesn't know which version to keep → you must fix it.

Steps:

1. Edit file and keep the correct line.

2. Mark as resolved:

```
git add file.py
```

```
git commit
```

Git Workflow Diagram

Git Workflow (Beginner Friendly)

